

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	257375	53.226470
1	[Git, Change, Create]	74452	15.397056
2	[Git, Change, Delete]	44503	9.203449
3	[Git, Change, Rename]	23284	4.815251
4	[File, Git]	15911	3.290476
5	[Git, Commit]	15582	3.222438
6	[Java, ByteCode, Parameter]	9227	1.908191
7	[Java, ByteCode, Member, Method]	8876	1.835602
8	[Java, ByteCode, Bound]	6300	1.302872
9	[Java, Value, ByteCode, Annotation]	5334	1.103099
10	[Java, ByteCode, Bound, ParameterizedType]	4849	1.002798
11	[Xml, Element]	2384	0.493023
12	[Java, ByteCode, Member, Field]	2187	0.452283
13	[Xml, Text]	1596	0.330061
14	[Java, ByteCode, Member, Constructor, Method]	1582	0.327166
15	[Java, ByteCode, Bound, WildcardType]	968	0.200187
16	[Java, ByteCode, Member, Method, Lambda]	851	0.175991
17	[Java, ByteCode, Bound, TypeVariable]	830	0.171648
18	[Type, File, Java, ByteCode, JavaType]	689	0.142489
19	[Type, File, Java, Class, ByteCode]	647	0.133803
20	[Json, Key]	533	0.110227
21	[Value, Json, Scalar]	523	0.108159
22	[Git, Change, Copy]	470	0.097198
23	[Type, File, Java, ByteCode, ResolvedDuplicate...	459	0.094924
24	[Java, ByteCode, GenericDeclaration, Member, M...	455	0.094096
25	[Java, Value, ByteCode, Primitive]	322	0.066591
26	[Author, Git, Person]	302	0.062455
27	[Type, File, Java, ByteCode, ExternalType]	301	0.062248
28	[Committer, Git, Person]	246	0.050874
29	[Maven, Dependency]	192	0.039707

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

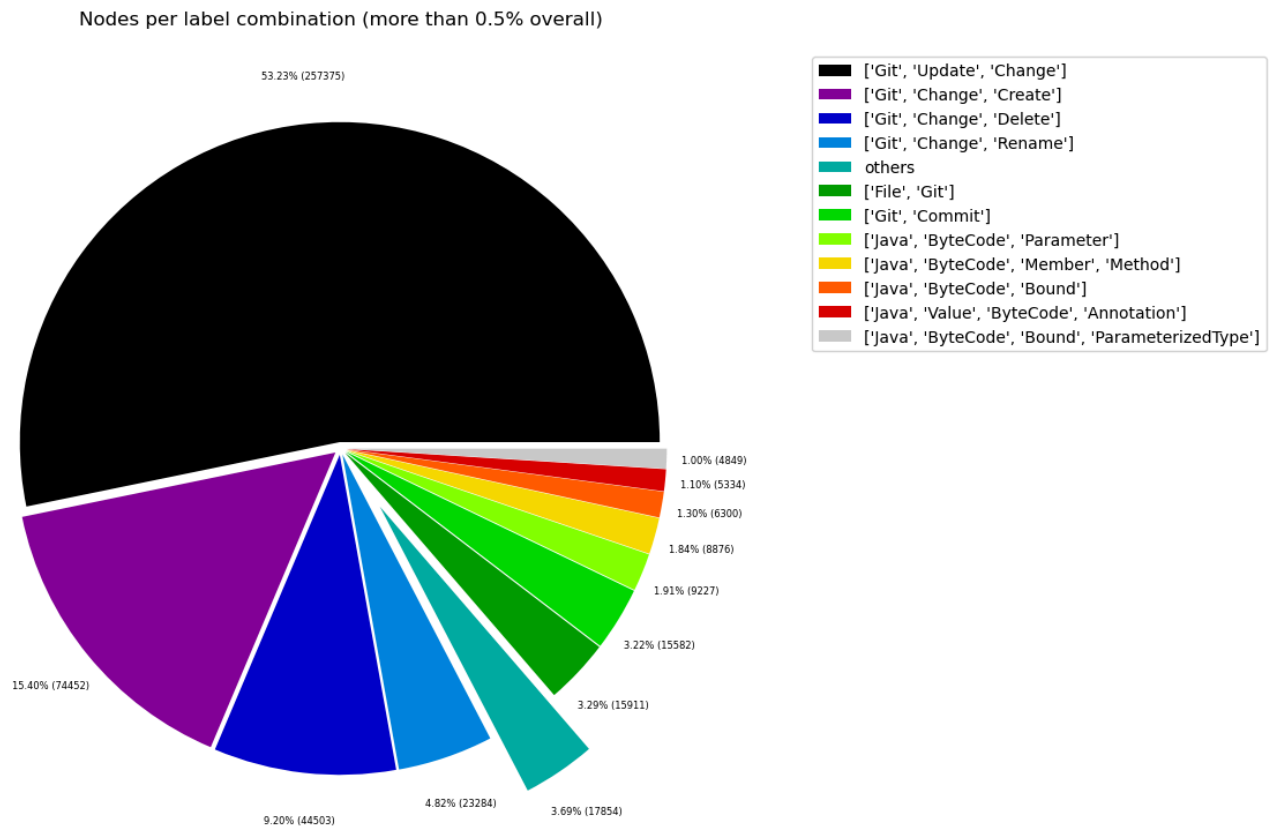


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000207
1	[Git, Branch]	1	0.000207
2	[Repository, File, Git]	1	0.000207
3	[File, Json]	2	0.000414
4	[File]	3	0.000620
5	[Maven, Exclusion]	4	0.000827
6	[Type, File, Java, ByteCode, GenericDeclaratio...	4	0.000827
7	[Value, Array, Json]	8	0.001654
8	[Type, File, Java, ByteCode, Throwable, Extern...	8	0.001654
9	[Java, ByteCode, GenericDeclaration, Member, C...	9	0.001861
10	[Maven, Scm]	11	0.002275
11	[File, Maven, Xml, Pom, Document]	11	0.002275
12	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002275
13	[Type, File, Java, ByteCode, Void]	11	0.002275
14	[Java, ManifestSection]	11	0.002275
15	[File, Java, Manifest]	11	0.002275
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002275
17	[File, Java, ServiceLoader]	12	0.002482
18	[Maven, PluginExecution]	13	0.002688
19	[File, Java, Properties]	13	0.002688
20	[Maven, ExecutionGoal]	13	0.002688
21	[Type, File, Java, ByteCode, Enum]	17	0.003516
22	[jQAssistant, Rule, Concept]	19	0.003929
23	[Maven, Configuration]	20	0.004136
24	[Maven, Plugin]	20	0.004136
25	[Xml, Attribute]	22	0.004550
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.008065
27	[Type, File, Java, ByteCode, Annotation]	42	0.008686
28	[Type, File, Java, Class, ByteCode, Throwable]	42	0.008686
29	[Xml, Namespace]	44	0.009099

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

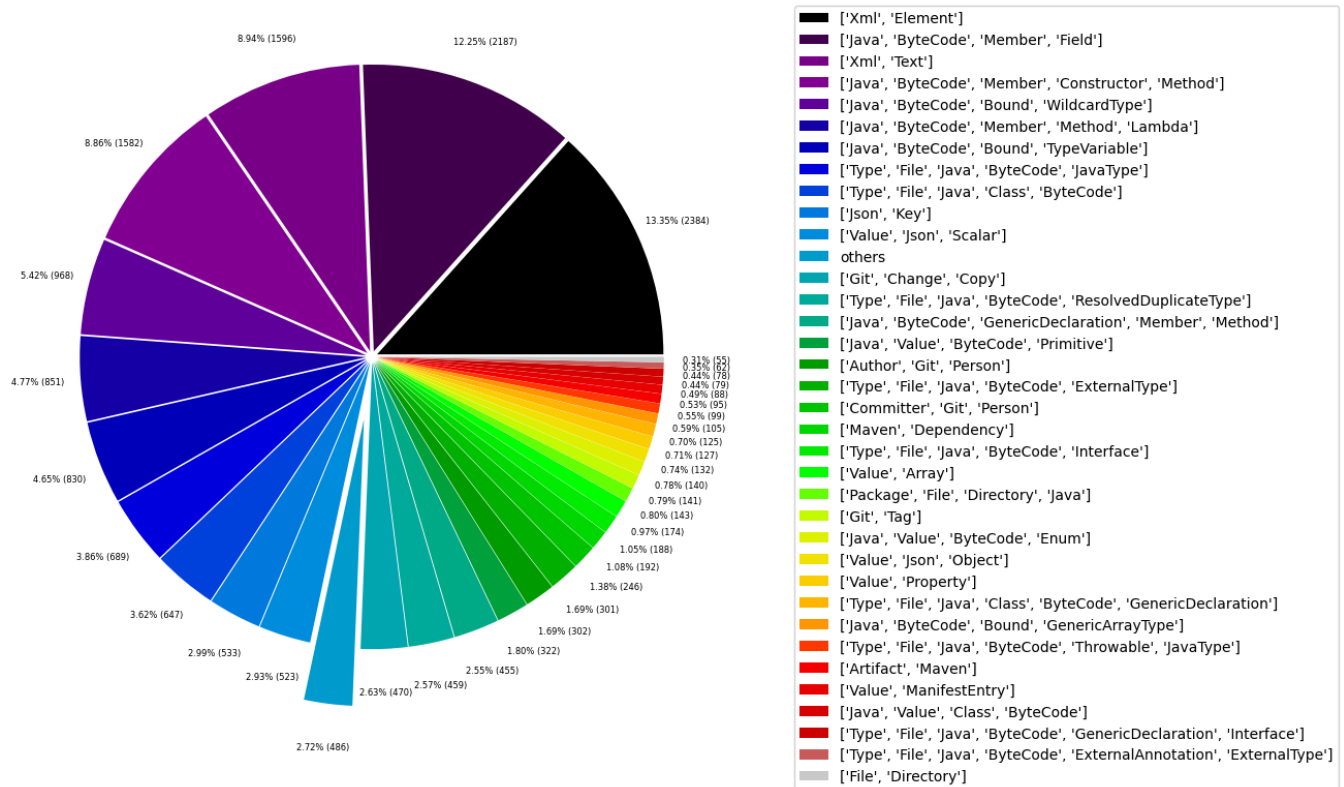


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

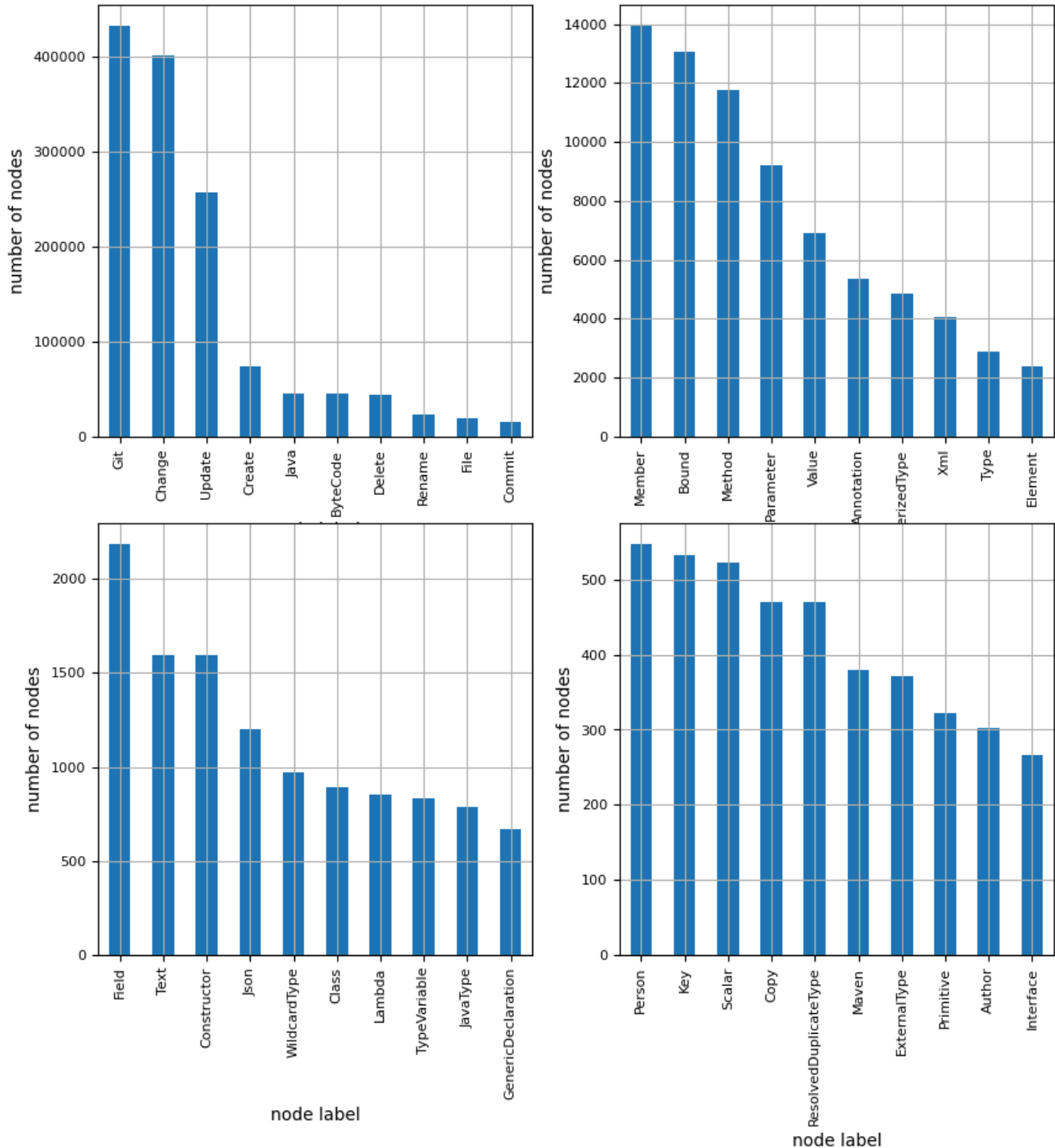
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	432268	89.395240
1	Change	400084	82.739423
2	Update	257375	53.226470
3	Create	74452	15.397056
4	Java	45189	9.345317
5	ByteCode	44988	9.303749
6	Delete	44503	9.203449
7	Rename	23284	4.815251
8	File	19047	3.939017
9	Commit	15582	3.222438
10	Member	13960	2.887000
11	Bound	13052	2.699221
12	Method	11773	2.434717
13	Parameter	9227	1.908191
14	Value	6927	1.432539
15	Annotation	5376	1.111784
16	ParameterizedType	4849	1.002798
17	Xml	4057	0.839008
18	Type	2874	0.594358
19	Element	2384	0.493023
20	Field	2187	0.452283
21	Text	1596	0.330061
22	Constructor	1591	0.329027
23	Json	1198	0.247753
24	WildcardType	968	0.200187
25	Class	893	0.184677
26	Lambda	851	0.175991
27	TypeVariable	830	0.171648
28	JavaType	788	0.162962
29	GenericDeclaration	671	0.138766
30	Person	548	0.113329
31	Key	533	0.110227
32	Scalar	523	0.108159
33	Copy	470	0.097198
34	ResolvedDuplicateType	470	0.097198
35	Maven	379	0.078379
36	ExternalType	371	0.076725
37	Primitive	322	0.066591
38	Author	302	0.062455
39	Interface	266	0.055010

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1501837

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	400084	26.639642
1	MODIFIES	400084	26.639642
2	UPDATES	257375	17.137346
3	CREATES	98206	6.539058
4	DELETES	67787	4.513606
5	COMMITTED	31164	2.075059
6	RENAMES	23284	1.550368
7	INVOKES	22195	1.477857
8	HAS_PARENT	18926	1.260190
9	OF_TYPE	18495	1.231492
10	DEPENDS_ON	17561	1.169301
11	HAS_FILE	15911	1.059436
12	HAS_COMMIT	15582	1.037529
13	DECLARES	14299	0.952101
14	OF_RAW_TYPE	13122	0.873730
15	HAS	9843	0.655397
16	HAS_NEW_NAME	9378	0.624435
17	RETURNS	8100	0.539339
18	READS	5929	0.394783
19	HAS_ACTUAL_TYPE_ARGUMENT	5753	0.383064
20	ANNOTATED_BY	5333	0.355098
21	OF_GENERIC_TYPE	4392	0.292442
22	CHANGED_TOGETHER_WITH	3686	0.245433
23	SIMILAR	3370	0.224392
24	CONTAINS	3183	0.211940
25	RETURNS_GENERIC	2586	0.172189
26	RESOLVES_TO	2568	0.170991
27	HAS_FIRST_CHILD	2384	0.158739
28	HAS_LAST_CHILD	2384	0.158739
29	HAS_ELEMENT	2362	0.157274

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

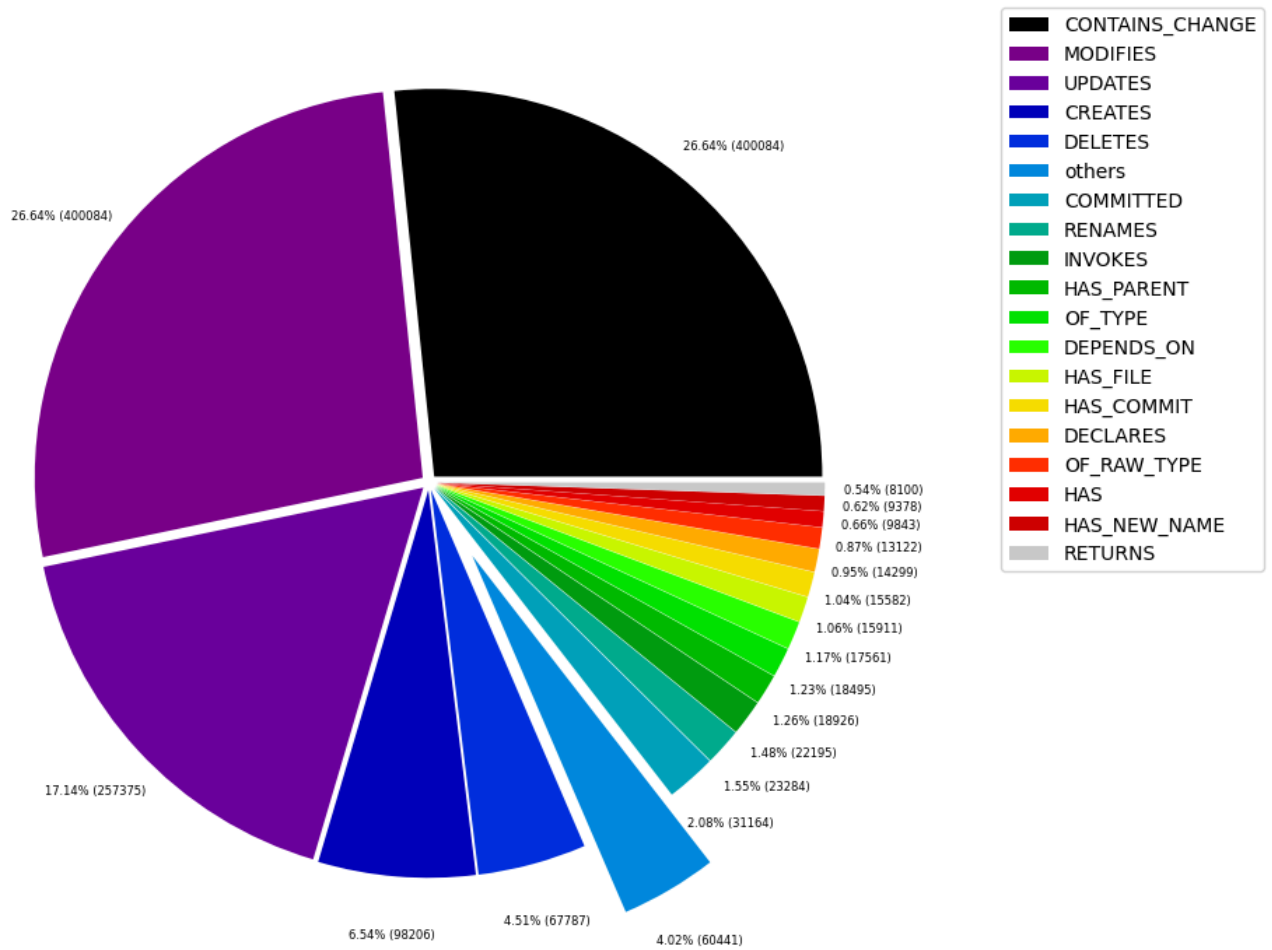


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

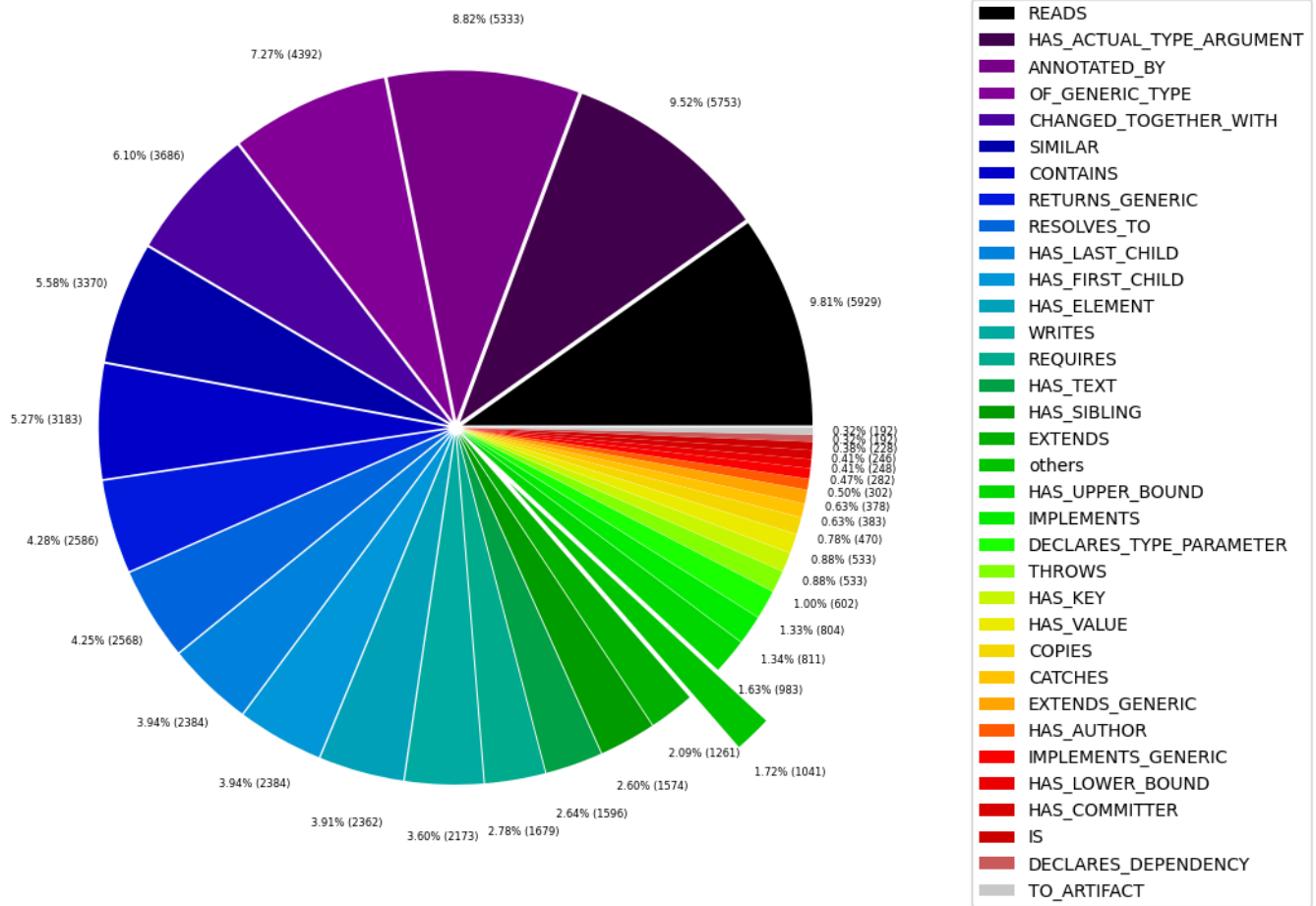
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000067
1	HAS_BRANCH	1	0.000067
2	HAS_HEAD	2	0.000133
3	THROWS_GENERIC	3	0.000200
4	EXCLUDES	4	0.000266
5	HAS_SCM	11	0.000732
6	DESCRIBES	11	0.000732
7	HAS_ROOT_ELEMENT	12	0.000799
8	HAS_GOAL	13	0.000866
9	HAS_EXECUTION	13	0.000866
10	INCLUDES_CONCEPT	19	0.001265
11	IS_ARTIFACT	20	0.001332
12	HAS_CONFIGURATION	20	0.001332
13	USES_PLUGIN	20	0.001332
14	OF_NAMESPACE	22	0.001465
15	HAS_ATTRIBUTE	22	0.001465
16	REQUIRES_TYPE_PARAMETER	26	0.001731
17	REQUIRES_CONCEPT	28	0.001864
18	DECLARES_NAMESPACE	44	0.002930
19	HAS_DEFAULT	53	0.003529
20	HAS_COMPONENT_TYPE	105	0.006991
21	CONTAINS_VALUE	128	0.008523
22	HAS_TAG	141	0.009389
23	ON_COMMIT	141	0.009389
24	COPY_OF	181	0.012052
25	TO_ARTIFACT	192	0.012784
26	DECLARES_DEPENDENCY	192	0.012784
27	IS	228	0.015181
28	HAS_COMMITTER	246	0.016380
29	HAS_LOWER_BOUND	248	0.016513

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberO
0	[Git, Update, Change]	MODIFIES	[File, Git]	257375		257375
1	[Git, Update, Change]	UPDATES	[File, Git]	257375		257375
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	257375		15582
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	74452		15582
4	[Git, Change, Create]	CREATES	[File, Git]	74452		74452
5	[Git, Change, Create]	MODIFIES	[File, Git]	74452		74452
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	44503		15582
7	[Git, Change, Delete]	MODIFIES	[File, Git]	44503		44503
8	[Git, Change, Delete]	DELETES	[File, Git]	44503		44503
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	23284		15582
10	[Git, Change, Rename]	CREATES	[File, Git]	23284		23284
11	[Git, Change, Rename]	RENAMES	[File, Git]	23284		23284
12	[Git, Change, Rename]	MODIFIES	[File, Git]	23284		23284
13	[Git, Change, Rename]	DELETES	[File, Git]	23284		23284
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	18915		15582
15	[Repository, File, Git]	HAS_FILE	[File, Git]	15911		1
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	15582		1
17	[Committer, Git, Person]	COMMITTED	[Git, Commit]	15582		246
18	[Author, Git, Person]	COMMITTED	[Git, Commit]	15582		302
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13220		8848
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9378		15911
21	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5323		8848
22	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5322		8848
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalAnnotatio...	4990		5334
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	4105		9227
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3797		9227
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3068		15911
27	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2799		4849
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2362		2384
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2256		6300

Graph Density

```
total_number_of_nodes (vertices): 483547
total_number_of_relationships (edges): 1501837
-> total directed graph density: 6.423124075366494e-06
-> total directed graph density in percent: 0.0006423124075366494
```